

A COVERING SYSTEM WITH MINIMUM MODULUS 44

GRAYSON MACKENZIE

ABSTRACT. We construct a covering system of the integers with pairwise distinct moduli whose least modulus is 44. Nielsen (2009) constructed such a system with least modulus 40, and Owens (2014) extended the method to 42. Starting from a variant of Owens’s construction, we extend it twice, first to least modulus 43 and then to 44. The new material consists of roots on fresh primes up to 211, small “device” families, and in-place repairs found by an exact counterexample loop. The final system consists of 618,413 families of congruences on 47 primes; after a standard finite truncation it is a finite covering system. Coverage was decided exactly, twice, by independent SAT encodings whose unsatisfiability proofs were checked by independent proof checkers.

1. INTRODUCTION

A *covering system* is a finite set of congruence classes $a_1 \bmod m_1, \dots, a_k \bmod m_k$ whose union is $\mathbb{Z}$. Erdős introduced covering systems with distinct moduli in 1950 [4]. He asked whether the least modulus of such a system can be arbitrarily large, a question he later described as perhaps his favourite problem. Hough [7] answered it in the negative, building on Filaseta, Ford, Konyagin, Pomerance and Yu [5]: every covering system with distinct moduli has least modulus at most 10^{16} . Balister, Bollobás, Morris, Sahasrabudhe and Tiba [1] gave a simpler proof and lowered the bound to 616,000.

Constructions lag far behind this bound. Successive records raised the least modulus to 25 (Gibson [6]) and to 40 (Nielsen [8]); see [8] for the history. Owens [9] extended Nielsen’s method to 42 in a master’s thesis.

Theorem 1. *There is a covering system of $\mathbb{Z}$ with pairwise distinct moduli whose least modulus is 44.*

Our contributions are as follows.

- (1) *The starting system* (§3). We work with UC85, a variant of Owens’s construction that differs from it in the placement of two carrier congruences and in the arrangement of four later sections. It has 356,364 families and least modulus 42.
- (2) *From 42 to 43 to 44* (§§4–5). We cut the unique modulus-42 family back to its deeper levels. We then re-cover exactly what it alone covered, using roots on fresh primes 89–173, a small device and in-place fills. We do the same for the unique modulus-43 family, with a 41-arrow in its slot, slot fills, roots on fresh primes 179–211 and 1,801 repairs found by an exact counterexample loop. Every modulus from 44 to 160 except 121 is used exactly once in the final system.
- (3) *Verification* (§6). Two independent exact checks establish coverage. Each builds a finite propositional formula that is satisfiable exactly when some integer other than a ray centre is uncovered (the single ray centre left is covered by the finite completion of §2); the formula is unsatisfiable, and every step of the solver’s refutation is checked by separate programs. Both checks also confirm distinct moduli.

The complete list of families and the certificates are available at <https://github.com/gr-mk/covering-minimum-modulus-44>.

2. FAMILIES, DISTINCT MODULI AND FINITE COMPLETION

Selectors and families. We describe congruence classes through *selectors* on single primes. For a prime p :

- a *fixed* selector $p: r \bmod p^k$ matches x when $x \equiv r \pmod{p^k}$;
- a *ray* selector $\text{ray}(p, [\ell, h], c, d)$, with $1 \leq \ell \leq h \leq \infty$, centre $c \in \mathbb{Z}$ and digit $d \in \{1, \dots, p-1\}$, matches $x \neq c$ when $e := 1 + v_p(x - c)$ lies in $[\ell, h]$ and $(x - c)/p^{e-1} \equiv d \pmod{p}$.

Equivalently, the ray matches the classes $x \equiv c + dp^{e-1} \pmod{p^e}$ for $\ell \leq e \leq h$. A *family* is a product of selectors on distinct primes, and it matches x when every selector does. Choosing one level e_p for each ray, a family is a union of single residue classes, one for each exponent vector, with modulus $\prod_p p^{e_p}$ (where $e_p = k$ for a fixed selector). These exponent vectors form the family's *exponent box*.

Distinct moduli. Every modulus of a family is divisible by exactly the primes of its support. Two families with different supports therefore never share a modulus, and the moduli of a system are pairwise distinct exactly when, among families with a common support, the exponent boxes are pairwise disjoint.

Ray centres and finite completion. A ray never matches its own centre. Consequently a system of families can cover every integer except finitely many *ray centres*. Our final system misses only the integer 0.

Rays are infinite recursions, and we make the system finite as Nielsen does [8, §2]. Each ray is truncated at a sufficiently deep level, and the finitely many deep classes that remain, including the ray centre 0, are covered by congruences that also involve an auxiliary prime P dividing no other modulus. Distinct powers of the prime of the ray (and of P where necessary) keep all moduli distinct, and every new modulus is a multiple of P , so it exceeds 44. We take $P = 223$, the least prime not used by the system.

3. THE STARTING SYSTEM

We use the notation of Nielsen [8] and Owens [9]. A q -arrow $q\uparrow(A_1, \dots, A_{q-1})$ covers a region by assigning to each nonzero q -adic digit j an *input* A_j , itself a set of congruences (with multipliers) that covers the region inside that digit's slot. Inputs of the form m (a bare multiplier) give single congruences whose modulus is q times m at the first level.

Owens's construction has the same shape as Nielsen's:

- a base on the primes 2, 3, 5 (§§3.1–3.3);
- sections for the primes 7 to 83 (§§3.4–3.20, the last three holding two primes each), whose arrows cover the parts of $\mathbb{Z}$ the earlier sections leave open.

The least modulus $42 = 2 \cdot 3 \cdot 7$ is attained once, by the family $2:0 \bmod 2 \cdot 3:1 \bmod 3\text{-ray}(7, [1, \infty), 0, 3)$ of §3.4.

The variant UC85. UC85 differs from Owens's construction [9] in three ways.

- (1) The modulus-55 congruence of §3.5, the bare input 1 of the sixth input $5\uparrow(1, 2, 3 \cdot 1, 4)$ of the 11-arrow, is placed at $1 \bmod 5$, as in Nielsen's construction.
- (2) The modulus-85 congruence of §3.7, the bare 1 in a $5\uparrow(1, 2, 4, 8\uparrow)$ input of its 17-arrow, is placed at $4 \bmod 5$ by permuting that 5-arrow to $5\uparrow(8\uparrow, 2, 4, 1)$. This gives §3.18's 67-arrow a carried 17-slot, and §3.18 then needs no 89-arrow.

- (3) The arrow layers of §§3.12, 3.14, 3.15 and 3.18 are arranged by an exact planner (a subset dynamic programme over the available input sets).

The result has 356,364 families on the 23 primes $2, \dots, 83$. Its least modulus is 42, attained once. It was certified by the method of §6.

4. FROM 42 TO 43

Let F_{42} be the modulus-42 family above. Changing its ray to $\text{ray}(7, [2, \infty), 0, 3)$ removes exactly the level-one class $10 \bmod 42$ and keeps the moduli $2 \cdot 3 \cdot 7^e$ for $e \geq 2$. Most of $10 \bmod 42$ is still covered by other families. The exact uncovered part U is 2.1% of the class, about $4.95 \cdot 10^{-4}$ of $\mathbb{Z}$, spread over 14 cells of the form (2-adic valuation, residue mod 5). We re-cover U with three ingredients:

- **A 7-device.** Seven families on $\text{ray}(7, [2, \infty), -4, j)$ inside $3 \bmod 7$, with moduli from 49 to 23,520. They cover fixed 7-digits of $x + 4$ exactly, and so act as exact carries for deeper covers.
- **In-place fills.** 604 families placed in empty slots of §§3.10, 3.12 and 3.15 (respectively 3, 75 and 526).
- **Seventeen roots on fresh primes** 89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167 and 173. Each root $q\uparrow$ covers one pinned piece of U with $q - 1$ inputs. The inputs are built from the other primes together with carries certified exactly: the device's digits, and root digits proved covered by an exact gate.

The resulting system has 540,826 families on 40 primes. Its least modulus is 43, attained once by the digit-1 input of §3.14's 43-arrow. The moduli $43, \dots, 48$ each occur once. Only F_{42} was changed and nothing was removed; this was checked independently.

5. FROM 43 TO 44

The unique modulus-43 family is the multiplier-1 input $\text{ray}(43, [1, \infty), 0, 1)$ of §3.14. We replace it by its tail $\text{ray}(43, [2, \infty), 0, 1)$, whose least modulus is 1,849, and re-cover what its first level alone covered.

- (1) **A 41-arrow in its slot.** In §3.14 the slot is filled by a 41-arrow built from the section's other inputs (177 families). §3.14's region then closes exactly.
- (2) **Slot fills.** The removed family also carried the 43-slot 1 of every inner 43-arrow elsewhere. We fill slot 1 of all 264 affected 43-arrow copies in §§3.15–3.19 (1,183 families) and add 2 families at one input of §3.20's 83-arrow.
- (3) **Exact census.** What is still uncovered lies in 1,282 pieces, all inside 22 root inputs of the 67- and 73-arrows of §§3.18–3.19. Their total measure is estimated, by sampling, at about 10^{-8} of $\mathbb{Z}$. The census is certified complete by a gate formula that is unsatisfiable with all pieces blocked (160 sub-cubes, all with checked proofs).
- (4) **Closing it.** Seven roots on the fresh primes 179, 181, 191, 193, 197, 199, 211 (74,424 families) cover the heavy inputs. 1,801 further families come from an exact counterexample loop: each uncovered point is traced to the input that misses it, and that input receives one thin family or a small local arrow, checked for collisions against the whole system.

The final system has 618,413 families on the 47 primes $2, \dots, 211$. Its least modulus is 44, attained once by $2:1 \bmod 4 \cdot \text{ray}(11, [1, \infty), 0, 1)$ of §3.5. Table 1 shows how densely the smallest moduli are packed.

TABLE 1. Moduli $m \leq 160$ of the final system. Each is used by exactly one family. The only integer in $[44, 160]$ that is not a modulus is 121.

range	moduli used
44–120	all 77 integers
121	not used
122–160	all 39 integers

6. VERIFICATION

Exact coverage. For each prime p of the system, the selectors on p partition $\mathbb{Z}$ into finitely many *local atoms*. These are residue classes modulo a fixed power of p , plus, for each ray, one atom per level up to a cap and one “deep” atom for the levels beyond it; a ray centre is its own limit atom. An integer’s atoms on the different primes determine which families contain it. Conversely, by the Chinese remainder theorem, every combination of atoms is realised by some integer. Coverage of $\mathbb{Z}$ outside the ray centres is therefore equivalent to the unsatisfiability of a propositional formula. That formula has one variable per atom, one exactly-one constraint per prime, and one clause per family forbidding its atoms.

For the final system the formula has 618,804 clauses. CaDiCaL [2] proves it unsatisfiable, and two further steps check that refutation:

- drat-trim [10] verifies the DRAT proof;
- the trimmed LRAT proof is replayed step by step as pure reverse unit propagation (77.7 million hints).

A second check shares no code with the first:

- It reads the system with inert stand-ins, so none of the builder’s code runs.
- It uses two separately written encoders. Their refutations were checked by drat-trim, by lrat-check, and by kissat [3] combined with drat-trim.
- Mutation controls confirm that neither encoder is vacuous. Removing a single needed root, repair or fill family makes the formula satisfiable, and the decoded model is an integer that only the removed family covers.

Distinct moduli and least modulus. The second check confirms distinct moduli in two ways:

- a sweep over exponent boxes within each support;
- a brute-force expansion of every modulus up to 10^{12} , with no repeats and none below 44.

Every family is well formed: its primes are sorted with one selector each, residues and digits are in range, and $\ell \geq 1$. A total of 50,912,769 sampled integers (not a proof) were all covered, except the ray centre 0.

7. A REMARK ON 45

The same method does not obviously reach 45. That would require removing the unique modulus-44 family, of §3.5, and covering again what it alone covers.

ACKNOWLEDGMENTS

The search and the certification described here were carried out with extensive assistance from Claude, an AI model developed by Anthropic, working under the author’s direction on the author’s computing resources. All results were checked by the exact methods of §6.

REFERENCES

- [1] P. Balister, B. Bollobás, R. Morris, J. Sahasrabudhe and M. Tiba, *On the Erdős covering problem: the density of the uncovered set*, Invent. Math. **228** (2022), 377–414.
- [2] A. Biere, K. Fazekas, M. Fleury and M. Heisinger, *CaDiCaL, Kissat, Paracooba, Plingeling and Treengeling entering the SAT Competition 2020*, Proc. SAT Competition 2020.
- [3] A. Biere and M. Fleury, *Gimsatul, IsaSAT and Kissat entering the SAT Competition 2022*, Proc. SAT Competition 2022.
- [4] P. Erdős, *On integers of the form $2^k + p$ and some related problems*, Summa Brasil. Math. **2** (1950), 113–123.
- [5] M. Filaseta, K. Ford, S. Konyagin, C. Pomerance and G. Yu, *Sieving by large integers and covering systems of congruences*, J. Amer. Math. Soc. **20** (2007), 495–517.
- [6] D. J. Gibson, *A covering system with least modulus 25*, Math. Comp. **78** (2009), 1127–1146.
- [7] R. Hough, *Solution of the minimum modulus problem for covering systems*, Ann. of Math. (2) **181** (2015), 361–382.
- [8] P. P. Nielsen, *A covering system whose smallest modulus is 40*, J. Number Theory **129** (2009), 640–666.
- [9] T. E. Owens, *A covering system with minimum modulus 42*, Master’s thesis, Brigham Young University, 2014.
- [10] N. Wetzler, M. J. H. Heule and W. A. Hunt Jr., *DRAT-trim: efficient checking and trimming using expressive clausal proofs*, in: SAT 2014, Lecture Notes in Comput. Sci. **8561**, Springer, 2014, 422–429.